

Ritabrata Chakraborty

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EDUCATION

University of Pennsylvania

M.S.E. in Robotics, GPA: ~ 4

Birla Institute of Technology and Science, Pilani (BITS Pilani)

B.E. in Mechanical Engineering — Minor in Data Science, CGPA: 8.57/10

Pennsylvania, USA

Aug '26 – May '28

Rajasthan, India

Oct '22 – May '26

PUBLICATIONS & PEER REVIEWS

Journal Publications

- R. Chakraborty and K. Kishore, “*Navigating Without Satellites: A Critical Survey of GNSS-Denied UAV Navigation.*” Submitted to **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026. [🔗](#)
- R. Chakraborty and K. Kishore, “*Multi-Critic Off-Policy Reinforcement Learning for Quadrotor Velocity Control in Unknown Indoor Environments.*” Manuscript in preparation for **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026.

Conference Publications

- Y. Wang, H. He, Y. Cao, J. Liang, R. Chakraborty, and G. Sartoretti, “*CogniPlan: Uncertainty-Guided Path Planning with Conditional Generative Layout Prediction.*” Accepted at **Conference on Robot Learning (CoRL)**, 2025; published in **Proceedings of Machine Learning Research (PMLR)**. [🔗](#)
- R. Chakraborty, T. Mian, and P. Kundu, “*An Efficient Approach for Synthetic Data Generation and Fault Diagnosis for Rotating Machinery.*” Accepted at **Prognostics and System Health Management Conference (PHM)**, 2025; published in **IET Conference Proceedings**. [🔗](#)
- K. Kishore and R. Chakraborty, “*Path Planning of Low-Altitude UAV for Tree Canopy Tracking and Orchard Monitoring.*” Publication awaiting intellectual property clearance, 2025. [🔗](#)

Poster Presentations

- R. Chakraborty, “*High-Velocity Impact Testing and Analysis of Layered Metal-Composite Shields Using Electromagnetic Railgun.*” Presented as poster at **Pulsed Power Science, Technology and Application (PPSTA)**, 2024.

Peer Reviews

- “*GS-SDE: 3D Gaussian Splatting-based Adaptive Scene Modeling in Semi-Dynamic Environments.*” Peer review completed for **IEEE Transactions on Automation Science and Engineering (T-ASE)**, 2026.
- “*Limits of Wearable Gait Analysis for Age Classification in Young Adults: Anthropometry Over Age.*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- “*Generative AI and Computer Vision Based Multi-Assessment System for Early Prediction and Diagnosis of Autism Spectrum Disorder (ASD).*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- “*AquaNode: Design and Implementation of a Low-Cost Automated Water Tank Monitoring and Valve Control System Using Arduino and Ultrasonic Sensing.*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- “*A research paper on IoT-Based Smart Irrigation Systems.*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- “*Hybrid AI-Based Autonomous Drone Swarm Navigation.*” Peer review completed for **IEEE Gujarat Section Conference (GSC)**, 2026.
- “*Autonomous Robot Navigation for Smart Indoor using A* and Deep Q-Network Model.*” Peer review completed for **IEEE Gujarat Section Conference (GSC)**, 2026.
- “*Deep Learning Driven Robot Simulation for Autonomous Navigation, Crop Health and Pesticide Spraying.*” Peer review completed for **IEEE Gujarat Section Conference (GSC)**, 2026.

RESEARCH EXPERIENCE

BITS Pilani - Research Intern

Supervisor: [Dr. Avinash Gautam](#), Associate Professor

- Literature study of vision-language grounded multi-robot coordination and VLMs-based navigation.

May '25 – Aug '25

CSIR-CEERI - Research Intern

Supervisor: [Dr. Kaushal Kishore](#), Principal Scientist

- Monocular vision-based UAV navigation for orchard monitoring using YOLOv11. [🔗](#)
- Multi-critic decomposition for off-policy reinforcement learning (TD3, SAC, DDPG) for autonomous indoor drone navigation. [🔗](#)

Jan '24 – May '26

MARMoT Lab, NUS - Research Intern

Supervisor: [Dr. Guillaume Sartoretti](#), Assistant Professor

- Uncertainty-guided path planning via conditional generative layout prediction (CogniPlan), benchmarked in the [CMU Exploration Environment](#). [🔗](#)
- Vision-attention-driven autonomous navigation with semantic understanding, extending [CogniPlan](#) and [ViNT](#).

Mar '25 – Dec '25

BITS Pilani - Research Assistant

Supervisor: [Dr. Mani Shankar Dasgupta](#), Associate Professor

- Physics-based launcher mechanical design and FEM analysis. [🔗](#)
- Computer vision pipeline for 3D trajectory reconstruction and shot classification using [PnLCalib](#).
- Finite element analysis of running track surfaces using hyperelastic material models.

Apr '24 – Oct '25

KU Leuven - Research Intern

Supervisor: [Dr. Pradeep Kundu](#), Assistant Professor

- Synthetic data generation for bearing fault diagnosis under data scarcity. [🔗](#)
- Physics-informed generative modeling for bearing fault diagnosis under data scarcity, validated on [LiteFormer](#).

Sep '24 – May '26

BITS Pilani - Research Assistant

Supervisor: [Dr. Rajneesh Choubisa](#)

- Primal world beliefs study surveying engineering undergraduates on core worldview dimensions. [🔗](#)

Aug '24 – Dec '24

BARC - Research Intern

Supervisor: [Dr. Rishi Verma](#), Head (ESDS), PPEMD

- Image-based visual servoing for automated radar control and UAV tracking using YOLOv8 and DeepSORT.
- Impact testing and analysis of metal-composite shields using electromagnetic railguns.

Jun '24 – Jul '24

INDUSTRY EXPERIENCE

Uber - ML Intern

Supervisors: [Mr. Siddarth Malreddy](#), Tech Lead Manager & [Mr. Ishan Nigam](#), Senior ML Engineer

- Object annotation automation, anomaly detection, and [TrackTrack](#)-based association checks in [uLabel](#).

Jul '25 – Dec '25

Knowledge Cultivator - Subject Matter Expert

Self-supervised

- Technical problem-solving and academic content authoring using MATLAB, CAD, and Python.

Nov '23 – Feb '24

PROJECTS

Vision-Guided Robotic Arm for Simulated Oral Cancer Screening	Aug '25 – Dec '25
Autonomous Drone Navigation MathWorks Global Student Drone Challenge 2025	Mar '25 – Apr '25
Air Quality Index Forecasting Using Time Series Models 🔗	Nov '24 – Nov '24
3D Lunar Surface Modeling from DEM Data ISRO Immersion Challenge 2024 🔗	Jul '24 – Jul '24
Cricket Team Optimization AMEX Campus Challenge 2024 (Product Track) 🔗	Jun '24 – Jul '24
ExoMy Rover Navigation and UR3 Arm Motion Planning ERC 2023 Remote 🔗	Apr '23 – Sep '23

TEACHING EXPERIENCE

Teaching Assistant	
CIS 4190/5190: Applied Machine Learning, University of Pennsylvania	Aug '26 – Dec '26
CS F320: Foundations of Data Science, BITS Pilani	Jan '26 – May '26
ME F218: Advanced Mechanics of Solids, BITS Pilani	Jan '25 – May '25
ME F216: Materials Science and Engineering, BITS Pilani	Sep '24 – Dec '24

HONOURS & SCHOLARSHIPS

Institute Merit-cum-Need Scholarship — Consecutive Recipient (Top 8% of Students)	Oct '22 – May '26
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AWARDS & ACHIEVEMENTS

Finalist — Meta PyTorch OpenEnv Hackathon (India) 🔗	Apr '26
3 rd Place — MathWorks Global Drone Student Challenge 2025	Mar '25
Finalist — ISRO Immersion Challenge (AI for Space & Geospatial Innovation) 🔗	Jul '24
Top 15 Overall, Top 5 in College — American Express Campus Challenge 2024 🔗	Jul '24
5 th Place & Best Maintenance Award — European Rover Challenge 2023 (Remote Finals)	Sep '23

LEADERSHIP EXPERIENCE

Deputy Director of Academic Programming	
Graduate and Professional Student Assembly (GAPSA), University of Pennsylvania	Aug '26 – Present
Vice-President	
South Asian Association (Rangoli), University of Pennsylvania	Aug '26 – Present
President & Secretary	
Mechanical Engineering Association (MEA), BITS Pilani	Jun '24 – May '26
President & Tech Fest Coordinator	
Indian Society of Heating, Refrigerating, and Air Conditioning Engineers (ISHRAE), BITS Pilani	Oct '24 – Jul '25
Project Manager	
Tinkerer's Lab (TL), BITS Pilani	May '24 – Jul '25
Treasurer	
Bengali Cultural Association (Moruchhaya), BITS Pilani	May '24 – Jul '25

VOLUNTEERING EXPERIENCE

Volunteer	
Junoon, National Service Scheme (NSS), BITS Pilani	Sep '23 – Sep '23

PROFESSIONAL ACTIVITIES

Graduate Student Member — IEEE	2026
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COMPETITIVE EXAMINATIONS

GRE: 317 (Quantitative: 168, Verbal: 149, AWA: 3)	2025
TOEFL iBT: 108 (Reading: 27, Listening: 30, Speaking: 23, Writing: 28)	2025
JEE Advanced: All India Rank 14372	2022

TECHNICAL SKILLS & COURSEWORK

Relevant Coursework: Machine Learning, Deep Learning, Machine Perception, Computer Vision, Foundations of Data Science, Applied Statistical Methods, Linear Algebra, Engineering Optimization, Differential Equations, Design of Machine Elements, Digital Twins
Programming Languages: Python, C++, C, C#, Java, Bash (Shell)
Robotics & Simulation: ROS (with Gazebo, Rviz), MAVROS, Navigation Stack, MoveIt, AirSim, Unity, Unreal Engine, MATLAB, Simulink, QGIS
AI & Data Science: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, OpenCV, Open3D, NumPy, Pandas, SciPy, StatsModels, Matplotlib, Seaborn, Weights & Biases (W&B), Excel
Cloud & MLOps: AWS (EC2, SageMaker), Azure (Foundry), Google Cloud Platform (GCP), Docker, Git, GitHub Actions
Hardware & Embedded Systems: NVIDIA Jetson (Nano, Orin), Raspberry Pi, Arduino, IMUs, Stereo Camera, 3D LiDAR
CAD & Mechanical Simulation: ANSYS Mechanical, SolidWorks, Fusion 360
Leadership & Communication: Event Planning, Team Management, Budget Allocation, Mentorship, Cross-Functional Coordination, Public Speaking, Stakeholder Outreach, Technical Writing
Productivity Software: Microsoft Office (Excel, PowerPoint, Word), Google Workspace